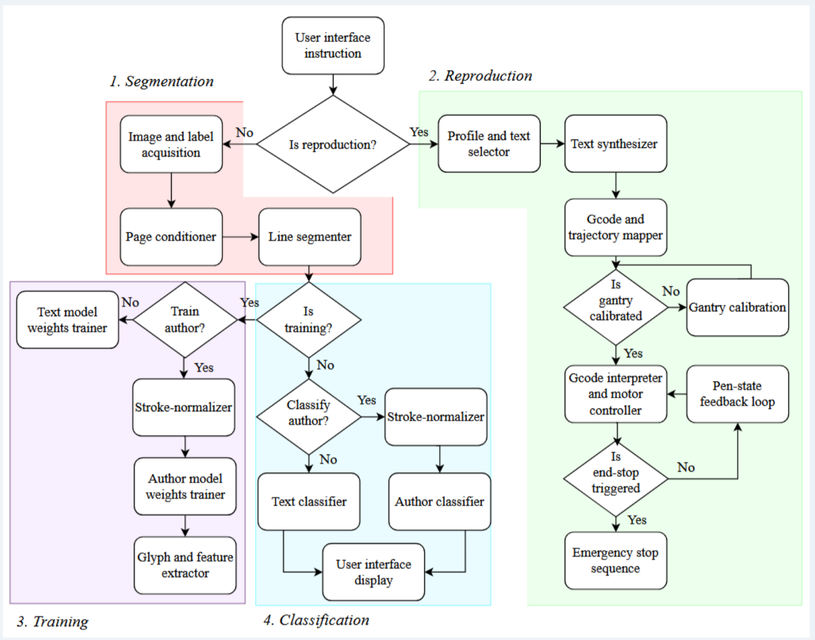
Handwriting Robot

Dylan Braune

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Department of Electrical, Electronic and Computer Engineering



Project overview

The system reads photographed handwriting and identifies the writer. It uses stored glyph shapes to reproduce new text in a selected style with a pen gantry. The system supports 11 writer styles.

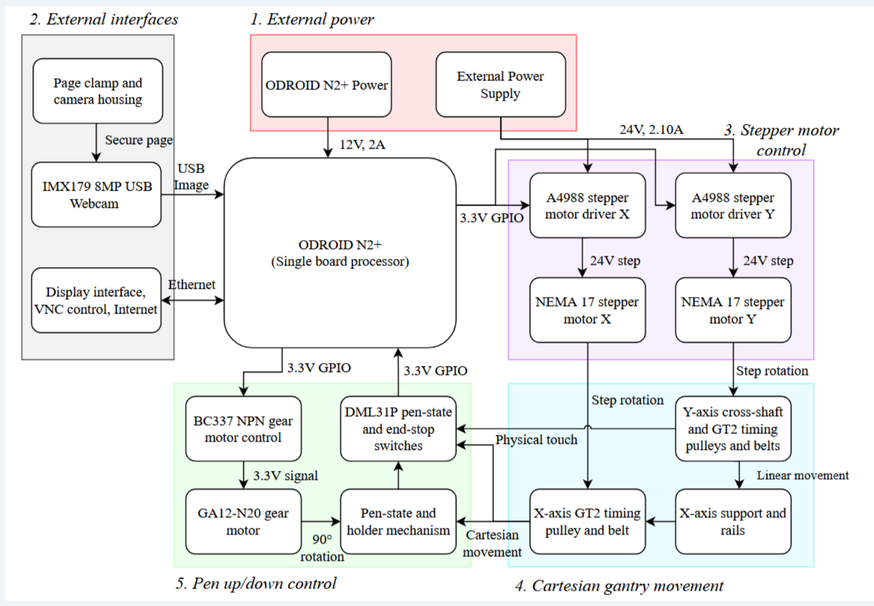
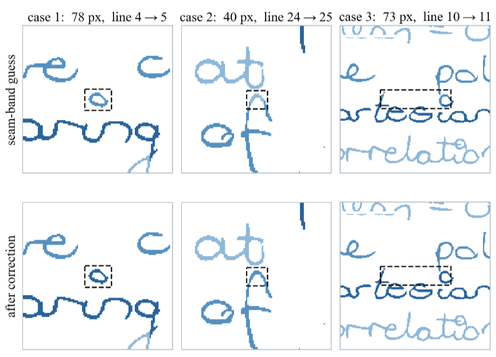
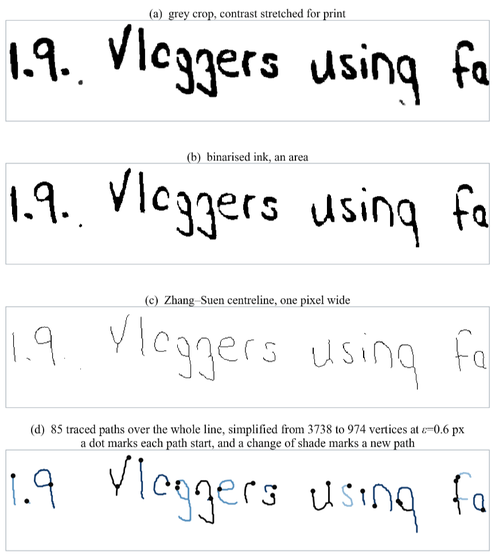


Image processing

Local background normalisation reduces shadows. Otsu's thresholding extracts the ink, and projection profiles with dynamic programming seams separate the lines. Zhang and Suen thinning recovers stroke centrelines.

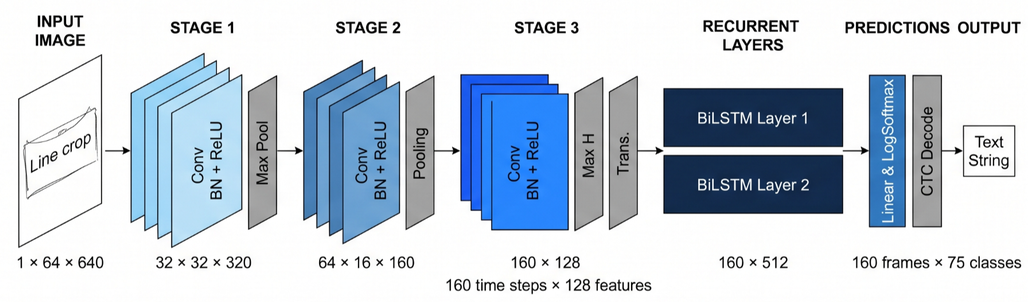




Recognition

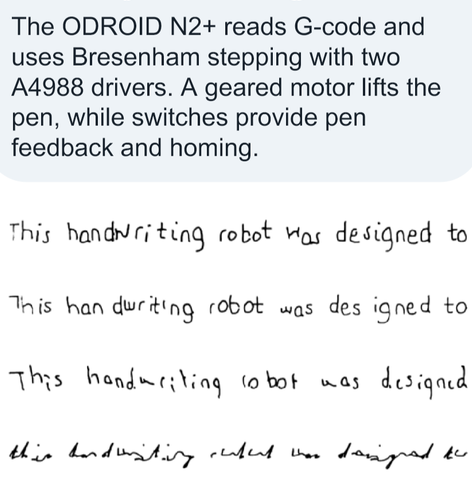
A CNN extracts line features. Two BiLSTM layers provide sequence context, and CTC decodes the text. A separate CNN identifies the writer from strokes normalised to 0.17 x-height.

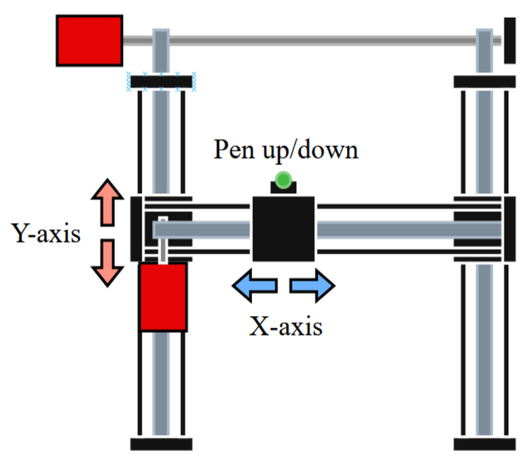
Parameter	Setting
Reader input	64 × 640 px
Writer styles	11
Stroke width	0.17 x-height



Gantry Plotting

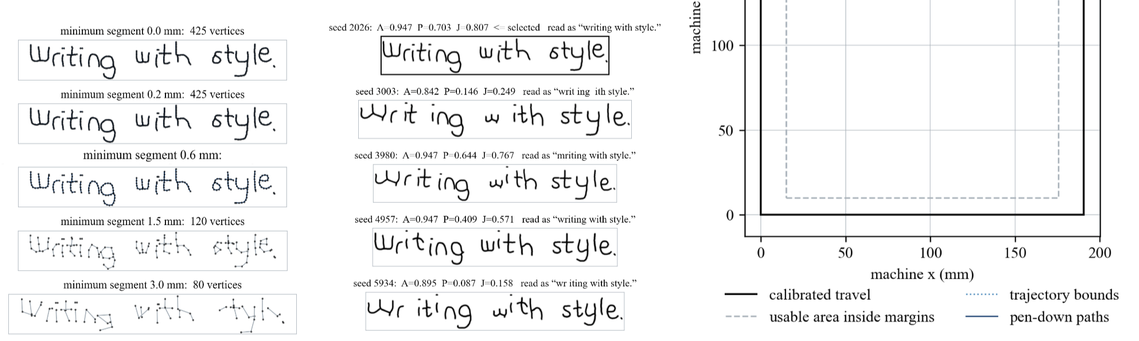
The ODROID N2+ reads G-code and uses Bresenham stepping with two A4988 drivers. A geared motor lifts the pen, while switches provide pen feedback and homing.





Handwriting synthesis

CTC forced alignment locates glyphs in labelled lines. Skeleton tracing stores their paths in writer profiles. The synthesiser selects variants, sets spacing and joins, then maps the paths to G-code.



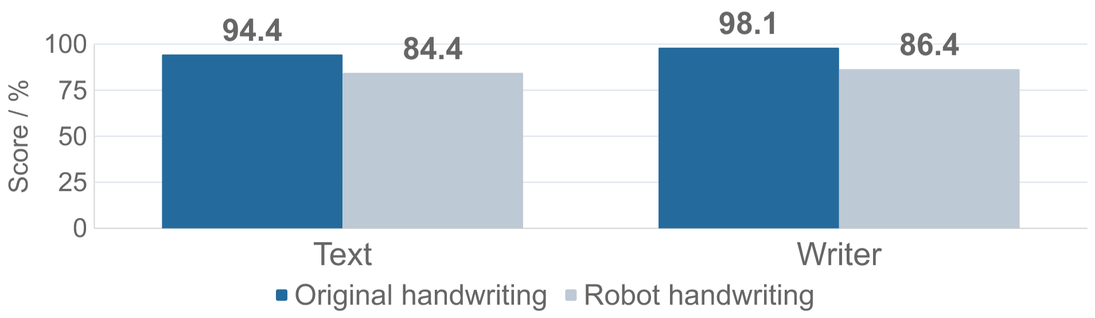
Measured performance

86.7%
relative shape score
against natural handwriting variation

0.297 mm
mean absolute length error

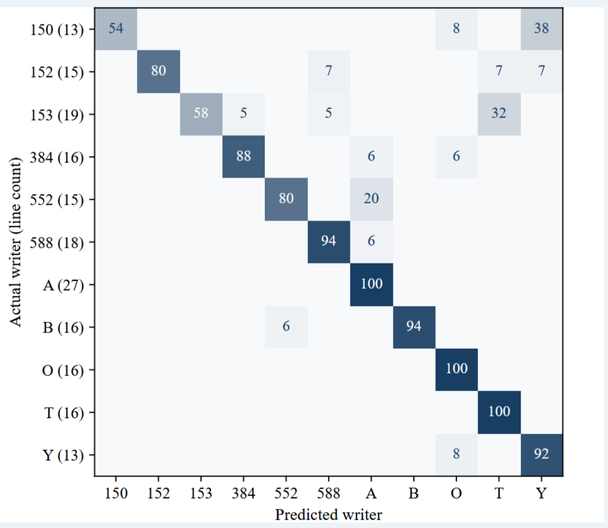
2.43 s
average completion time per character, 25 jobs

Recognition scores



Original validation and robot output are separate test sets.
Classification time remains above target.

Robot writer identification



UNIVERSITEIT VAN PRETORIA
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YUNIBESITHI YA PRETORIA

Faculty of Engineering, Built Environment and Information Technology

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